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The influences of boundary spanning capabilities and long-term orientation on opportunism in inter-firm relationships

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ABSTRACT

Opportunism is a common issue in inter – firm relationships, which increases transaction costs, impairs firm performance and supply chain efficiency, and deteriorates relationship quality. Previous studies have intensively investigated factors influencing opportunism but overfocusing on the organisational level. Recently, supply chain researchers have called for identifying factors influencing opportunism at the individual level to gain a more comprehensive understanding of opportunism. Based on transaction cost theory, resource dependence theory, and boundary spanning theory, this study seeks to advance the emerging research stream on opportunism at the individual level by investigating how buyer agents' boundary spanning capabilities and firms' long – term relationship orientation can help manage supplier agents' opportunism. Using a cross – sectional data set of 406 buyer agents from manufacturing companies collected from February to April 2021 in Japan, we tested a set of hypotheses through regression and conditional process analyses. Our findings suggest that buyer agents' strategic communication and firms' long – term relationship orientation are critical factors that reduce supplier agents' opportunism.

KEYWORDS

Inter-firm relationship; opportunism; boundary spanning capability; long-term orientation; Japan

Introduction

In today's global competition and uncertainty, business companies face numerous issues, including opportunistic behaviour or opportunism in inter – firm relationships (Han et al. 2022; Wang, Huo, and Tian 2021). Opportunism is defined as 'self – interest seeking with guile' (e.g. lying, cheating, distorting facts) by human actors in business relationships (Williamson 1985, 47). Opportunism increases transaction costs (Dahlstrom and Nygaard 1999), impairs firm performance and relationship quality (Williamson 1985), and hinders supply chain efficiency (Daultani et al. 2015). Thus, the reduction of opportunism in inter – firm relationships has drawn considerable attention from researchers and practitioners.

Previous studies on inter – firm relationships have primarily examined factors influencing organisational – level opportunism, where opportunism is attributed to and managed by companies (Bhattacharya, Singh, and Nand 2015; Han et al. 2022; Handley and Benton 2012; Huo et al. 2018; Parkhe 1993; Wang et al. 2016; Yang et al. 2021). However, an emerging trend focuses on individual opportunism, highlighting the role of buyer agents as boundary spanners of buyer firms in managing suppliers' opportunistic behaviours during business transactions (Gelderman, Semeijn, and Verhappen

2020; Hawkins, Lewin, and Amos 2012; Hawkins, Pohlen, and Prybutok 2013; Srinivasan and Srivastava 2012; Tangpong, Hung, and Ro 2010).

To effectively manage business relationships with suppliers and mitigate their opportunistic behaviours, buyer agents must possess strong boundary spanning capabilities (BSCs), which include purchasing knowledge, communication skills, and compromise ability (Zhang, Viswanathan, and Henke 2011). These BSCs are particularly important during the pre – contract stage, where buyer agents utilise them to analyse relevant information and negotiate contract terms with suppliers. This information includes product details (design, quality, alternatives), delivery specifications (just – in-time, schedule), cost factors (raw materials, production, taxes, exchange rates), and supplier capabilities (alternative options, reputation) (Akrouf and Diallo 2017). In this stage, buyers are vulnerable to suppliers' opportunistic behaviours, such as inflating costs. BSCs help buyer agents mitigate these risks (Zhang, Viswanathan, and Henke 2011).

Furthermore, BSCs continue to be crucial in the post – contract stage for buyer agents to control suppliers' opportunistic behaviours. For instance, when there are sudden increases in raw material prices, significant changes in exchange rates, or disruptions in product delivery due to adverse weather conditions,

suppliers often attempt to renegotiate terms in their favour. In such situations, buyer agents' purchasing knowledge and skills are vital in understanding the situation and developing negotiation strategies to adjust contracts with suppliers, thereby avoiding disadvantages for the buyer company (Dowell, Morrison, and Heffernan 2015; Pan and Choi 2016). However, individual opportunism in inter – firm relationships has been overlooked in previous literature (Pham, Yokozawa, and Anh 2022; Tangpong, Hung, and Ro 2010). Individual opportunism is critical to consider because individual agents – i.e. buyer and supplier agents – are primarily responsible for conducting and managing business transactions; they have many opportunities to perform opportunistically (Hawkins, Lewin, and Amos 2012). In many cases, opportunistic behaviours in business exchanges that damage firm performance and cooperation between firms are intentionally performed by individual agents rather than companies (Lyons, Krachenberg, and Henke 1990). Thus, this study is a part of the emerging trend focusing on individual opportunism.

Additionally, only a limited number of studies have empirically operationalised and tested the role of boundary spanning capabilities (BSCs) in managing opportunism. transaction cost theory (TCT) and resource dependence theory (RDT) highlight the significance of knowledge limitations regarding products, markets, competitors, and communication skills as major factors hindering individual agents' effective control over their counterparts' opportunistic behaviours (Williamson 1985). Moreover, knowledge is considered a crucial resource that enables a party to exert influence and shape the behaviours of other parties within supply chains (Pfeffer and Salancik, 1978). However, there have been few empirical studies that have operationalised and tested the propositions put forth by TCT and RDT.

On the other hand, existing research on boundary spanning theory (BST) has already operationalised relevant knowledge and skills for buyer agents involved in business exchanges through the concept of BSCs. These BSCs encompass professional purchasing knowledge, compromise ability, and strategic communication, which have been shown to play a significant role in facilitating business exchanges and sustaining inter – firm relationships (Vesalainen, Rajala, and Wincent 2020; Zhang, Viswanathan, and Henke 2011; Zhang, Wu, and Henke 2015). Hence, this study aims to investigate the role of buyer agents' BSCs in managing supplier agent opportunism.

Moreover, there is a lack of studies that have elaborated on the integrated framework of individual opportunism developed by previous research, which emphasised the effects of organisational factors as contextual moderating conditions that can enhance or reduce the role of individual factors in controlling

individual opportunism (Gelderman, Semeijn, and Verhappen 2020; Hawkins, Pohlen, and Prybutok 2013). Therefore, this study aims to address this gap by considering the role of long – term orientation (LTO) between buyer and supplier firms as an organisational contingency factor that could potentially enhance the impact of buyer agents' BSCs on supplier agents' opportunism. Numerous studies have confirmed the role of LTO in inter – firm relationships, such as reducing transaction costs (Williamson 1985), facilitating information exchanges (Dyer and Chu 2011; Lee et al. 2018), fostering trust (Chung, Sternquist, and Chen 2006), promoting collaboration (Yunus and Kurniawan 2015), and creating a cooperative environment (Gundlach, Achrol, and Mentzer 1995; Lui and Ngo 2012) among business partners. Thus, to enrich relevant theories and to provide practical implications, we address the following research question: To what extent do buyer agents' BSCs and LTO between buyer and supplier firms impact supplier agent opportunism and how?

To address the research question, this study collected data from a survey by 406 purchasing agents in Japanese manufacturing companies. Conditional process analysis is applied, which is 'an analytical strategy that integrates mediation and moderation analysis intending to examine and test hypotheses about how mechanisms vary as a function of context or individual differences' (Hayes and Rockwood 2020, 1). In this study, we test the main effects of buyer agents' boundary spanning capabilities on supplier agents' opportunism and test the moderating effects of how the main effects vary in the context of long – term orientation between buyer and supplier companies.

This study makes four potential contributions. First, we extend TCT, RDT, and the literature on factors influencing opportunism in inter – firm relationships by empirically examining opportunism at the individual level. Second, our study enriches BST by empirically verifying the influence of buyer agents' BSCs in controlling opportunistic behaviours from supplier agents. Third, we also contribute to the emerging research trend on individual opportunism by considering LTO between buyer and supplier firms as an organisational factor that may enhance the effectiveness of buyer agents' BSCs in controlling supplier agents' opportunism. Fourth, we also enrich our understanding of how LTO is important in Japanese culture by testing the proposed relationships in Japanese manufacturing companies.

The remainder of this paper is structured as follows. In the next section, we provide a literature review about factors influencing individual opportunism in inter – firm relationships and the role of buyer agents' BSCs in managing inter – firm relationships. Next, we

explain how we developed the hypotheses, explaining how buyer agents can effectively use their BSCs and firms' LTO to mitigate supplier agent opportunism. Next, we address the methodology and results. Following are the conclusions where we present theoretical contributions and implications for practitioners. The final section is about limitations and future research.

Literature review

Factors influencing individual opportunism in inter – firm relationships

Both individual and organisational factors have an impact on individual opportunism. In terms of individual factors, the cooperativeness and honesty of individual agents have been found to influence their engagement in opportunistic behaviours (Hawkins, Lewin, and Amos 2012; Hawkins, Pohlen, and Prybutok 2013; Tangpong, Hung, and Ro 2010). Moreover, individual agents' perceptions of their counterparts, including perceiving them as undermining their honour or demonstrating opportunistic behaviours, can positively influence their decision to engage in opportunistic behaviours in response (Gelderman, Semeijn, and Verhappen 2020).

On the other hand, organisational factors also play a significant role. Relational norms between buyer and supplier companies have been found to have a negative influence on the occurrence of opportunistic behaviours among their agents (Tangpong, Hung, and Ro 2010). Additionally, the power dynamics between buyer and supplier companies can impact individual agents' opportunism, with agents from more powerful companies more likely to engage in opportunistic behaviours towards agents from less powerful companies (Hawkins, Pohlen, and Prybutok 2013). Furthermore, the organisational environment created by companies can also affect individual agents' opportunism. For instance, a workplace that promotes cooperative ethical values can reduce the likelihood of agents behaving opportunistically (Hawkins, Pohlen, and Prybutok 2013). Conversely, the conduct of managers and leaders, such as engaging in opportunistic leadership or displaying wilful ignorance, can increase agents' opportunism (Hawkins, Lewin, and Amos 2012; Hawkins, Pohlen, and Prybutok 2013).

Furthermore, the interaction effects between individual and organisational factors play a crucial role in managing individual agents' opportunism. Relational norms between buyer and supplier firms alone cannot effectively reduce individual agents' opportunism unless those agents exhibit cooperativeness (Tangpong, Hung, and Ro 2010). Similarly, the presence of honesty and integrity among individual

agents may not be sufficient to decrease their likelihood of engaging in opportunistic behaviours if their managers and leaders display high levels of opportunistic behaviours or if the company does not prioritise ethical conduct (Hawkins, Lewin, and Amos 2012; Hawkins, Pohlen, and Prybutok 2013). The interaction between individual and organisational factors highlights the complexity of managing opportunism and emphasises the importance of considering both sets of factors simultaneously.

Role of buyer agents' BSCs in managing inter – firm relationships

The importance of individual agents' knowledge and skills in facilitating inter – firm relationships and performance has also been recognised in BST (Doney and Cannon 1997). According to BST, individual agents who represent their firms in business exchanges with partner firms are referred to as boundary spanners. In the context of business exchanges with supplier firms, buyer agents serve as typical examples of boundary spanners and play crucial roles in fostering relationships with individuals from supplier firms, as well as improving the overall performance of the supply chain (Adams 1976; Aldrich and Herker 1977; Doney and Cannon 1997). These buyer agents utilise their knowledge and skills to bridge the gap between their own organisation and the supplier firms, ensuring effective communication, collaboration, and relationship management.

There are three BSCs that buyer agents need to effectively carry out business transactions with suppliers: (1) strategic communication capability: The effectiveness of buyer agents in presenting their firm's strategic intention and goals to suppliers (Ireland and Webb 2007; Perrone, Zaheer, and McEvily 2003); (2) professional knowledge: The knowledge possessed by buyer agents that enables them to competently carry out their job responsibilities (Doney and Cannon 1997); and (3) ability to reach compromise with suppliers: The effectiveness of buyer agents in mediating and overcoming differences that arise between their firms and the suppliers they interact with (Friedman and Podolny 1992).

Previous literature has examined the effectiveness of buyer agents' BSCs in building trust between them and supplier agents and between buyer firms and supplier firms (Doney and Cannon 1997; MacDuffie and Helper 2006; Zhang, Viswanathan, and Henke 2011; Zhang, Zheng, and Li 2019). Buyer agents with a high level of strategic communication skills and knowledge can increase suppliers' willingness to invest in their firms (Zhang, Wu, and Henke 2015) and supplier embeddedness (Cai et al. 2021). Professional

knowledge helps buyer agents understand and integrate the knowledge shared by suppliers to reinforce cooperation values (Teigland and Wasko 2003). BSCs are critical for buyer agents to identify and attract the best suppliers, implement coordination processes, enhance internal and external interactions, and develop tight bonds (Tchokogu   and Merminod 2021). The summary of the research on the impacts of BSCs in managing inter – firm relationships is presented in Table 1.

In sum, the literature review highlights three main points. First, individual opportunism is influenced by both individual and organisational factors, in which organisational factors are conditional factors that strengthen or weaken the role of individual factors in reducing individual opportunism. Second, there is a lack of empirical studies examining the role of buyer agents' BSCs in managing supplier agents' opportunism. Third, BST investigates buyer agents' knowledge and skills in inter – firm relationships and operationalises their knowledge and skills using three types of BSCs: professional purchasing knowledge, compromise ability, and strategic communication. While these capabilities are crucial in managing supplier relationships, there is limited understanding of how they can effectively control supplier agents' opportunism. Therefore, it is necessary to empirically investigate the impact of buyer agents' BSCs on supplier agents' opportunism to elaborate on the fundamental propositions of TCT and RDT and to enrich knowledge of the roles of buyer agents' BSCs in BST.

Hypotheses development

Impact of buyer agents' professional purchasing knowledge on supplier agents' opportunism

Buyer agents can leverage their professional knowledge to effectively manage their working relationships with supplier agents. Firstly, a strong knowledge foundation allows buyer agents to negotiate favourable contracts with suppliers as they are well – informed about crucial terms and conditions (Bals et al. 2019). Additionally, buyer agents with comprehensive product knowledge can accurately assess product value and production costs, making it difficult for supplier agents to engage in opportunistic behaviours (Revilla and Knoppen 2015).

Secondly, a high level of knowledge equips buyer agents with the ability to negotiate effectively in unexpected situations. Buyer agents' knowledge is crucial in obtaining a clear understanding of the circumstances, enabling them to avoid any disadvantages for their companies (Revilla and Knoppen 2015).

Thirdly, a strong knowledge base empowers buyer agents to establish cooperative ties with suppliers. In inter – firm relationships, exchanges of knowledge and skills are often integral components. Therefore, supplier agents are more likely to cooperate and collaborate with knowledgeable buyer agents, as they recognise the opportunity to learn and benefit from their expertise (Barbat 2011).

Overall, buyer agents' professional knowledge plays a vital role in contract negotiations, managing unforeseen circumstances, and fostering cooperative relationships with suppliers, thereby reducing the likelihood of opportunistic behaviours.

Hypothesis 1. Buyer agents' professional purchasing knowledge reduces supplier agents' opportunism.

Impact of buyer agents' compromise ability on supplier agents' opportunism

Compromise ability is a crucial attribute for buyer agents in effectively managing supplier agents' opportunistic behaviours. As intermediaries between buyer and supplier firms, buyer agents play a pivotal role in establishing and maintaining cooperative relationships (Vesalainen, Rajala, and Wincent 2020). Compromise ability enables buyer agents to reduce conflicts and address interest mismatches that may arise between themselves and supplier agents, as well as between the buyer and supplier firms (Zhang, Viswanathan, and Henke 2011). By mitigating conflicts and interest discrepancies, business exchanges can operate smoothly, leading to increased satisfaction among supplier agents and others within supplier firms, thereby reducing the likelihood of opportunistic behaviours (Zhang, Viswanathan, and Henke 2011).

Moreover, buyer agents with strong compromise ability can foster an environment where supplier agents feel understood and valued. This sense of understanding and care enhances trust and respect between buyer agents and supplier agents, creating a conducive atmosphere for cooperation and reducing the occurrence of opportunistic behaviours (Cai and Yang 2008). When supplier agents perceive that their interests and concerns are acknowledged and taken into account by buyer agents, they are more motivated to engage in collaborative and trustworthy behaviours.

In summary, compromise ability is a critical attribute for buyer agents in managing supplier agents' opportunistic behaviours. It helps in conflict resolution, addressing interest mismatches, fostering satisfaction,

Table 1. Summary of research on the impact of BSCs in managing inter – firm relationships.

Authors	Research type	Methodology and sample	Inter – firm relationship performance	Effects of BSCs on inter – firm relationship performance
Au and Fukuda (2002)	Empirical	- Questionnaire survey administered to 232 expatriates in Japan and English – speaking countries - Regression analysis	- Expatriates' role ambiguity - Expatriates' role benefits - Expatriates' use of community resources - Expatriates' power - Boundary spanners' job satisfaction	BSCs help expatriates gain more role benefits, utilise community resources, and reduce feelings of ambiguity. In addition, BSCs help boundary spanners gain more power and job satisfaction in their own companies.
Teigland and Wasko (2003)	Empirical	- Questionnaire survey administered to 83 information technology services and consulting companies in Europe - Structural equation modelling	- Creativity - Integrating knowledge	Boundary spanning communication positively influences creativity and boundary spanners' performance in integrating knowledge across organisational boundaries.
Zhang et al. (2011)	Empirical	- Questionnaire survey administered to 345 buyers in the US automotive industry and 143 buyers in the European food industry - Structural equation modelling analysis	- Suppliers' trust in agents - Suppliers' trust in firms	Buyer agents' BSCs, in the form of strategic communication with supplier firms, help increase trust between buyer and supplier firms, while buyer agents' professional purchasing knowledge and compromise ability help reinforce supplier firms' trust in buyer agents.
Zhang et al. (2015)	Empirical	- Questionnaire survey administered to 198 buyers in the US and 181 buyers in Europe - Structural equation modelling analysis	- Communication - Trustworthiness - Willingness to invest	BSCs, in terms of professional knowledge and strategic communication, help enhance communication quality between spanners' firms and supplier firms, which in turn increases spanners' firms' trustworthiness and supplier firms' willingness to invest.
Huang et al. (2016)	Empirical	- Questionnaire survey administered to 225 manufacturer – distributor dyads in China - Structural equation modelling analysis	- Relationship quality - Relationship trust - Relationship satisfaction - Relationship commitment	Boundary spanning roles have significant influences on inter – organisational relationships. The effectiveness of boundary spanners in processing information and external representation helps boost communication and cooperation and reduce conflict between organisations. This improves relationship quality, trust, satisfaction, and commitment in inter – organisational relationships.
Cai et al. (2017)	Empirical	- Questionnaire survey administered to 277 phone manufacturers in China - Path analysis	- Inter – firm collaboration - Inter – firm conflict	Boundary spanners' interactions and capabilities positively impact the development of interpersonal ties among boundary spanners, which helps strengthen inter – firm collaboration and reduce inter – firm conflict.
Schotter et al. (2017)	Conceptual model	Meta – analysis literature review and conceptual model development	- Operational diversity: global activities, global value chain, inter – form ecosystem - Organisational capabilities: system effectiveness, process improvement, cost efficiencies	A sense of honour and identity, strategic decision – making, and the abilities, skills, and experiences of individual boundary spanners are critical to boosting the operational diversity and capabilities of organisations in the global supply chain.
Dekker et al. (2019)	Empirical	- Questionnaire survey administered to 200 firms in French - Process regression analysis	- Collaboration scope - Firm performance	BSCs help significantly improve firm performance in business exchanges in terms of on – time delivery, offer quality, innovation, and profit margins. Further, boundary spanners' effectiveness when performing their jobs in business exchanges helps enhance the positive impact of narrow – scope collaborations in firm performance.
Son et al. (2019)	Empirical	- Questionnaire survey administered to 344 purchasing agents in Korean manufacturing firms - Hierarchical regression analysis	Relationship performance	Boundary spanners need to be able to act as mediators between their focal firms and partner firms. The capacity to influence one's counterparts in information, decision, profit, and risk sharing is critical to facilitating collaborative supply chain relationships.
Zhang et al. (2019)	Empirical	- Questionnaire survey administered to 287 manufacturing firms in China - Two – stage least squares regression	Trust restoration	Individual boundary spanners have important roles in repairing relationships in business exchanges after conflicts and opportunism.
Collien (2021)	Conceptual	Meta – analysis	- Power - Knowledge sharing - Cross – cultural collaboration	Spanning capabilities such as communication, knowledge, and compromise ability are critical for individual spanners to gain power and to effectively use that power to drive knowledge sharing and cross – cultural collaboration.

(Continued)

Table 1. (Continued).

Authors	Research type	Methodology and sample	Inter – firm relationship performance	Effects of BSCs on inter – firm relationship performance
Vesalainen et al. (2020)	Empirical	- Mixed – methods design: Questionnaire survey administered to 349 individual purchasers in Finnish manufacturing companies and follow – up interviews with 20 purchasers. - Cluster analysis for the survey data and cross – case analysis for the interview data	Purchaser agents' persuasive orientations (relational, authoritarian, and competitive)	Purchaser agents, as boundary spanners in business exchanges, need to have a high level of spanning capabilities to gain relational, authoritarian, and competitive advantages when working with suppliers.
Cai et al. (2021)	Empirical	- Questionnaire survey administered to 225 purchasing managers in China - Path analysis	- Counterparts' trust in spanners - Personal closeness with their counterparts - Counterparts' firms' trust in boundary spanners' firms	Individual spanners with a high level of BSCs have close personal ties with their counterparts. This helps deepen trust between themselves and their counterparts, as well as between their own and their counterparts' firms.

and building trust, thereby reducing the likelihood of opportunistic behaviours in inter – firm relationships.

Hypothesis 2: Buyer agents' compromise ability reduces supplier agents' opportunism.

Impact of buyer agents' strategic communication on supplier agents' opportunism

Strategic communication plays a crucial role for buyer agents in effectively managing supplier agents' opportunistic behaviours. Buyer agents with strong strategic communication capabilities are able to effectively communicate buyer firms' purchasing strategies and plans to supplier firms, facilitating a better understanding of buyer firms by supplier agents and others within supplier firms (Makaoui and Aloui 2015; Zhang, Wu, and Henke 2015). For instance, buyer agents engage in communication with supplier agents regarding buyer firms' plans to expand or streamline product lines, or their desire to introduce changes to product designs. Such information is valuable for suppliers to plan their manufacturing schedules, production processes, and technology investments in order to establish long – term cooperation with the buyers (Zhang, Viswanathan, and Henke 2011). These collaborative planning efforts foster closer working relationships between suppliers and buyers, reducing the likelihood of opportunistic behaviours (Zhang, Viswanathan, and Henke 2011).

Furthermore, effective communication helps to build trust between supplier and buyer agents, creating a positive environment where supplier agents appreciate the relationship and are less inclined to engage in opportunistic behaviours (Zhang, Zheng, and Li 2019). By maintaining open and transparent lines of communication, buyer agents can establish a sense of trust and mutual understanding with

supplier agents, leading to a stronger commitment to the relationship and a reduced intention to act opportunistically.

In summary, strategic communication is critical for buyer agents in effectively managing supplier agents' opportunistic behaviours. By effectively conveying purchasing strategies and plans, fostering collaborative planning efforts, and building trust, buyer agents can establish cooperative relationships and mitigate the likelihood of opportunistic behaviours in inter – firm exchanges.

Hypothesis 3: Buyer agents' strategic communication capability reduces supplier agents' opportunism.

Moderating role of LTO

Indeed, the opportunism of individual agents is not solely influenced by individual factors, but also by moderating effects from the strength of inter – firm relationships (Gelderman, Semeijn, and Verhappen 2020; Hawkins, Lewin, and Amos 2012; Hawkins, Pohlen, and Prybutok 2013; Tangpong, Hung, and Ro 2010). In this study, we consider LTO between buyer and supplier firms as a crucial contextual factor that enhances the effectiveness of buyer agents' Balanced Scorecards (BSCs) in managing supplier agents' opportunism.

In a business relationship, LTO refers to the desire and value that an economic actor places on establishing a long – term bond with a specific exchange partner (Ryu, Park, and Min 2007). When LTO is high, buyer and supplier firms consider each other as essential partners and engage in more frequent and strategic collaborations. This may involve spending more time discussing manufacturing plans, product development strategies, and innovative ideas (Lee et al. 2018). As a result, buyer agents operating in such conditions are

in a favourable business climate to work closely with supplier agents and effectively communicate and compromise with both parties.

The presence of a strong LTO facilitates the development of a cooperative and collaborative environment, promoting open communication, mutual understanding, and trust between buyer and supplier agents. Buyer agents with high LTO are more likely to engage in long – term relationship building and engage in behaviours that reduce opportunistic tendencies among supplier agents. The strength of the buyer – supplier relationship, influenced by LTO, amplifies the impact of buyer agents' BSCs in managing and mitigating supplier agents' opportunism.

Hypothesis 4a: LTO between buyer and supplier firms enhances the role of buyer agents' professional knowledge in reducing supplier agents' opportunism.

Hypothesis 4b: LTO between buyer and supplier firms enhances the role of buyer agents' compromise ability in reducing supplier agents' opportunism.

Hypothesis 4c: LTO between buyer and supplier firms enhances the role of buyer agents' strategic communication capability in reducing supplier agents' opportunism.

The hypotheses are illustrated in Figure 1.

Research methodology

Sampling and data collection

This study collected data by surveying Japanese manufacturing companies, where LTO is a crucial competitive advantage (Dyer and Chu 2003, 2011; Sako and Helper 1998). By developing long – term relationships with business partners, companies can enhance supply chain integration and collaboration (Dyer and Chu 2011), improve communication and information sharing (Chung, Sternquist, and Chen 2006), improve the

manufacturing process (Clark and Fujimoto 1991), and increase human and capital assets (Aoki and Wilhelm 2017). Hence, in the context of Japanese manufacturing companies, examining how LTO between firms can help (as a contextual factor) individual buyer agents effectively use their BSCs to control supplier agent opportunism is necessary.

The target participants in the survey were purchasing staff in the sales and planning sections of manufacturing companies in Japan responsible for business activities with suppliers. As purchasing staff generally represent buyer firms in business transactions with suppliers' sales staff, they are important in managing suppliers' opportunism (Zhang, Viswanathan, and Henke 2011). Further, a purchasing staff member usually works with more than one sale staff member from one supplier firm. Thus, we asked purchasing staff to consider the supplier company with the highest transaction value they were dealing with and the person from the supplier company they contacted most frequently.

To recruit participants, we cooperated with GMO Research (<https://gmo-research.com/>), a professional research company. They offer a platform accessed by over 40 million individuals. When participating in research, individuals are awarded incentive points, redeemable for gift cards, cash, or merchandise.

Data collection proceeded in two stages. The first entailed a pre – screening survey to recruit participants who matched our criteria. Qualified participants needed to work in Japanese manufacturing companies' sales and planning sections. They had to work or have worked as buyer agents in inter – firm business transactions. In the second stage, GMO Research opened its survey platform for selected participants. The company required participants to log in to the survey platform using their member ID verification code; thus, each participant could complete the survey only once (Schoenherr, Ellram, and Tate 2015). Further, the question items were presented in random order, and we provided reverse questions to

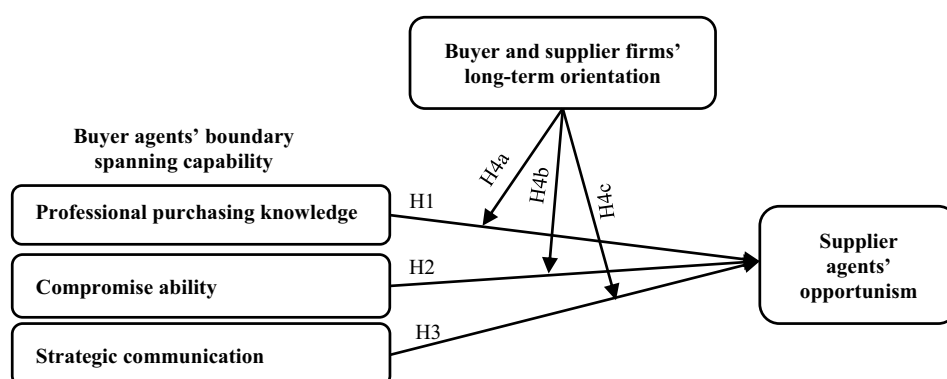


Figure 1. Conceptual model.

prevent unengaged responses and ensure objectivity (ibid.).

Data were collected from February to April 2021. There were 413 samples, and after screening to remove outliers, we obtained a dataset of 406 usable samples, which reached the rule of thumb for the sample size for a moderation model (Nunnally 1978). The sample profile is provided in the [Appendix](#).

Scale measurement

We adapted instruments from the literature to measure constructs in the conceptual model. Opportunism scales included 10 items adapted from Carson et al. (2006), Gundlach et al. (1995), and Heide et al. (2007), measuring the extent to which a buyer agent perceives that a supplier agent displays opportunistic behaviours. The sample items were 'This agent violated contractual terms for their own benefit', 'This agent violated informal agreements for their own benefit', and 'This agent altered facts to drive a better deal'.

Buyer agents' BSC measures were adapted from Zhang et al. (2011). Professional purchasing knowledge was measured with four items indicating the extent to which a buyer agent has knowledge about products, suppliers' capability, competitors, and the market. Compromise ability was measured with three items indicating buyer agents' ability to reach equitable solutions for both sides, considering suppliers' interests. The three items of strategic communication capability measured a buyer agent's effectiveness in conveying their firm's purchasing plans and strategies to the supplier firm and explaining the buyer firm's suggestions regarding the programme/contract or agreements to the supplier firm. Sample items were 'I have knowledge about the supplier firm's capabilities (e.g. quality, timely delivery, safety standards)', 'I generally strive to reach solutions that are equitable to both the supplier firm and my firm when differences arise', and 'I am effective at explaining my firm's purchasing strategy to the supplier firm'.

Six items adapted from Chen et al. (2011) were used to measure LTO. These items measured the extent to which a buyer firm has the intention and willingness to maintain and develop a long-term relationship with a supplier firm. Sample items were 'We focus on long-term goals in this relationship' and 'Maintaining a long-term relationship with this supplier is important to us'.

The items were translated from English to Japanese and then back-translated to determine the accuracy of the translation (Huo et al. 2018). Respondents were required to evaluate the extent to which they agreed with the statements on

a five-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree).

Results

Reliability and validity

Reliability

We used SPSS and AMOS software to analyse the data. We used the following procedure suggested by Narasimhan and Jayaram (1998). First, confirmatory and exploratory factor analyses were used to test the reliability of measurement (Gorsuch 1983). We generated standard factor loadings for each measurement item showing the percentage of the variable explained by the item, and if an item had a standard factor loading less than 0.5, it was removed. Then, we calculated the Kaiser-Meyer-Olkin (KMO) value to measure the sampling adequacy for each variable. Composite reliability (CR) and Cronbach's α were used to measure the internal consistency of scale items. The KMO value for compromise ability was above 0.6, while the KMO value for all other scales was above 0.8. The CR for all constructs was above 0.7. As the rule of thumb for KMO and CR values is 0.6 and 0.7, respectively, the sampling was adequate, or the proportion of variance may have been caused by high underlying factors (Nunnally 1978). The Cronbach's α of all scales was greater than 0.7, demonstrating the internal consistency and reliability of the items and constructs (Hair et al. 1995). In addition, correlations for all pairs of variables were calculated. [Table 2](#) presents the correlation matrix, CR, Cronbach's α , and KMO values.

Validity

A model fit index was used to test the validity of the model, that is, the fits between the observed data and a particular probability distribution (Schumacker and Lomax 2010). First, the chi-square test with χ^2 and χ^2/df indicators was conducted to compare the observed data and expected results. The results showed that $\chi^2/\text{df}=2.342$, which was less than 5, indicating a good benchmark. Next, the overall fit indices for model evaluation were calculated, including the Comparative Fit Index (CFI), Normed Fit Index (NFI), Incremental Fit Index (IFI), Tucker-Lewis Index (TLI), and Relative Fit Index (RFI). The results showed that CFI=0.913, NFI=0.858, IFI=0.913, TLI=0.905, and RFI=0.845. Values above 0.8 indicate a good model fit (Hu and Bentler 1999); all indicators in this study ranged from 0 to 1. Finally, the Root Mean Square Error of Approximation (RMSEA) and Standardised Root Mean Square Residual (SRMR) were used to assess how far from perfect our model was. The indicators should range from 0 to

Table 2. Correlations, reliability, and validity.

Constructs	1.	2.	3.	4.	5.
1. Opportunism	1				
2. Strategic communication ability	-0.203***	1			
3. Long-term orientation	-0.287***	0.727***	1		
4. Professional knowledge	-0.152**	0.948***	0.672***	1	
5. Compromise ability	-0.165**	0.963***	0.742***	0.911***	1
Cronbach's α	0.938	0.838	0.885	0.854	0.719
CR	0.938	0.838	0.888	0.861	0.714
KMO	0.946	0.680	0.895	0.858	0.680

$N = 406$; ** $p < 0.010$; *** $p < 0.001$; CR = composite reliability; KMO = Kaiser – Meyer–Olkin value.

Table 3. Results of linear regression and conditional process analysis.

Hypothesis	Direct effects	Coefficient β	Evaluation
H1	Professional knowledge $\rightarrow$ Opportunism	1.968***	Not supported
H2	Ability to reach compromise $\rightarrow$ Opportunism	7.933***	Not supported
H3	Strategic communication $\rightarrow$ Opportunism	-7.548***	Supported
Hypothesis	Moderating effects of LTO	Coefficient β	Evaluation
H4a	Professional knowledge $\rightarrow$ Opportunism	-0.078*	Partially supported
H4b	Compromise ability $\rightarrow$ Opportunism	-0.084**	Partially supported
H4c	Strategic communication $\rightarrow$ Opportunism	-0.153*	Supported

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

1; higher values reflect poorer fits. Our data showed that RMSEA = 0.058 and SRMR = 0.0697, both less than 0.07; thus, our model had an acceptable fit level (Cho et al. 2020).

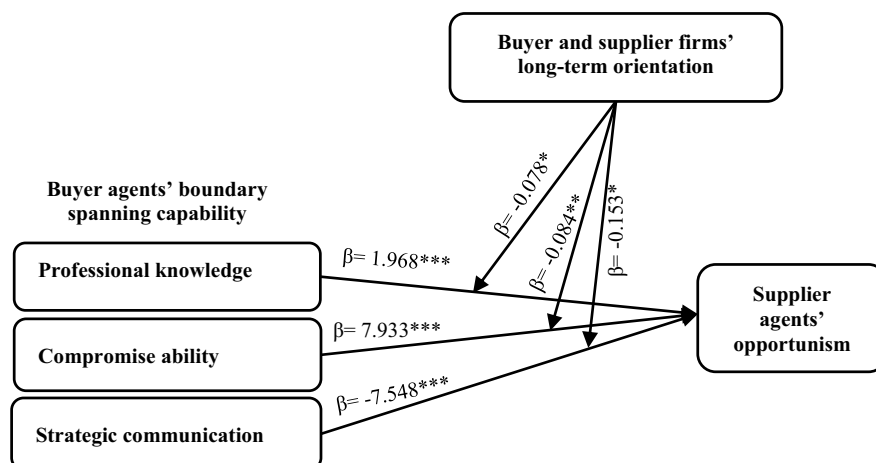
Hypotheses testing

After analysing reliability and validity, linear regression and conditional process analyses were used to test the direct and moderating effects, respectively (Hayes 2017). The analytical results are presented in Table 3 and Figure 2 below.

Regarding main effects, Hypothesis 1 suggested that buyer agents' professional knowledge reduces supplier agents' opportunism, but our results showed a positive relationship ($\beta = 1.968$; $p < 0.001$). This effect

can occur in some situations. For example, buyer agents are knowledgeable about raw materials and product values. Therefore, if products are not made from the best raw materials, but suppliers still want to sell at a high price, they would lie about the materials to get their desired price.

Similarly, Hypothesis 2, which suggested that buyer agents' compromise ability reduces supplier agents' opportunism ($\beta = 7.933$; $p < 0.001$), was not supported. One possible explanation for this unexpected outcome is that in business exchange negotiations, it is assumed that suppliers always act out of self – interest, whereby they want to sell their product and benefit as much as they can (Williamson 1985). Thus, if buyer agents act as middlemen and try to reach a compromise regarding suppliers' requirements, they may inadvertently create

**Figure 2.** Data analysis results.

more favourable conditions that the suppliers can then exploit.

Hypothesis 3 implies that buyer agents' strategic communication reduces supplier agents' opportunism. Our findings strongly support Hypothesis 3 ($\beta = -7.548$; $p < 0.001$). The results align with previous studies emphasising the importance of strategic communication in managing healthy inter-firm relationships (Zhang, Viswanathan, and Henke 2011; Zhang, Wu, and Henke 2015; Zhang, Zheng, and Li 2019). A buyer agent's strategic communication capability provides a supplier agent with information about the buyer firm's future purchasing plans and strategies. This enables the supplier agent and others from their firm to understand the buyer firm and determine the necessary preparation for cooperation (Ireland and Webb 2007). In other words, buyer agents with good strategic communication capabilities will be able to effectively communicate with suppliers and foster close cooperation with supplier agents and between their firms and supplier firms, helping reduce supplier agents' opportunistic behaviours.

Regarding interaction effects, Hypothesis 4a, which proposed that LTO between a buyer firm and a supplier firm enhances the role of buyer agents' professional knowledge in reducing supplier agents' opportunism, was partially supported as LTO changed the effect of professional knowledge on opportunism from positive to negative ($\beta = -0.078$; $p < 0.05$). Similarly, our findings partially supported Hypothesis 4b because the data analysis implied that LTO between a buyer firm and a supplier firm changed the role of buyer agents' compromise ability from increasing to reducing supplier agents' opportunism ($\beta = -0.084$; $p < 0.01$). However, Hypothesis 4c was supported ($\beta = -0.153$; $p < 0.05$), which means the role of buyer agents' strategic communication capability in reducing

supplier agents' opportunism will be stronger if the buyer and supplier firms have a long-term relationship orientation.

The moderation effects aligned with the literature (Lee et al. 2018; Ryu, Park, and Min 2007). LTO is not only necessary for business relationship management between buyer and supplier companies but also for the relationships between their buyer agents and supplier agents (Hakansson 1982) because LTO between the two companies will create a working atmosphere for their agents that then helps shape the relationship between them (Hawkins, Lewin, and Amos 2012). When a buyer company expresses the intention for long-term cooperation, a supplier agent understands that the relationship with the buyer agent will be ongoing, which helps the supplier agent feel secure; as such, the supplier agent will be less likely to act opportunistically (Hawkins, Pohlen, and Prybutok 2013).

In addition, Figure 3 uses graphs to show the interaction effects between LTO and professional knowledge, compromise ability, and strategic communication capability. In Figure 3a,b,c the dependent variable opportunism is placed on the vertical axis, with the independent variables – professional knowledge, compromise ability, and strategic communication capability – on the horizontal axis. In each figure, two graph lines are displayed; the solid line represents the low LTO case, and the dashed line represents the high LTO case.

In Figure 3a, the solid line is above the dashed line, which means that for the same level of professional knowledge, the high LTO case will result in lower opportunism than the low LTO case (Aiken, West, and Reno 1991). Figure 3b shows that the solid line lies above the dashed line; thus, with the same levels of compromise ability, the high LTO case will lead to

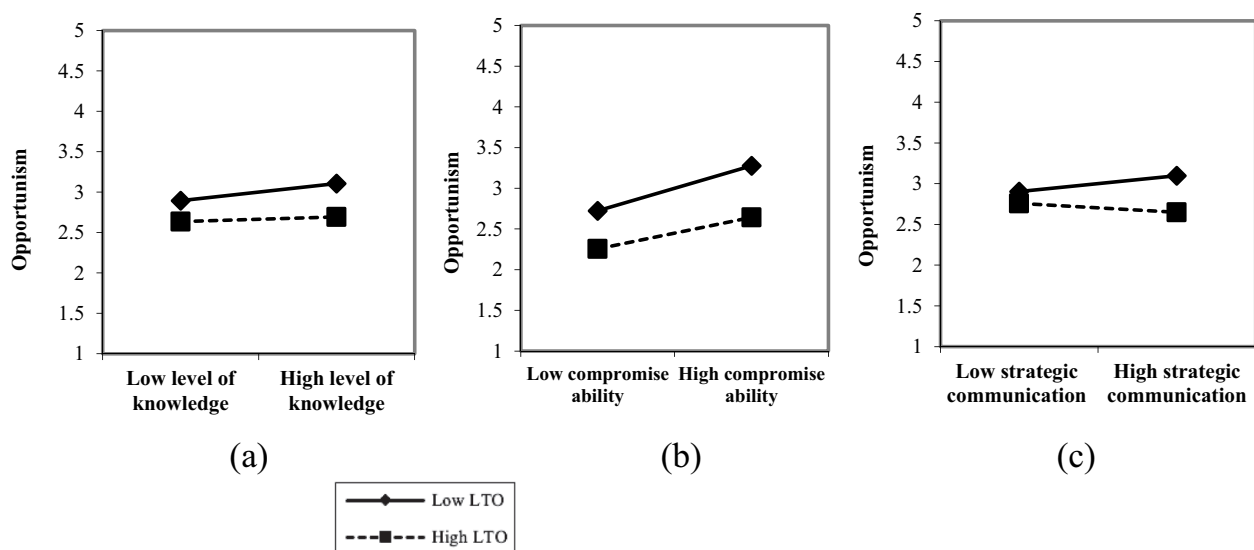


Figure 3. Moderating effects of LTO in the relationship between: (3a) opportunism and level of knowledge. (3b) opportunism and compromise ability. (3c) opportunism and strategic communication.

lower opportunism than the low LTO case (Aiken, West, and Reno 1991).

In Figure 3c, the solid line (low LTO) is also above the dashed line (high LTO), which means that with the same levels of strategic communication, if there is a high LTO, the level of opportunism will be lower than the low LTO (Aiken, West, and Reno 1991). Moreover, the graphs on Figure 3c show that the relationship between buyer agents' strategic communication and opportunism is negative in the high LTO but positive in the low LTO. The result, even more emphasises the role of LTO in motivating strategic communication's effectiveness in controlling opportunism.

Prior studies have shown that when communicating with business partners, companies need to consider carefully why (not) sharing information and knowledge and what information and knowledge to share with whom because that will affect companies' competitive advantage, interdependence, and bargaining power in the business relationships (Kembro and Näslund 2014; Kembro, Selviaridis, and Näslund 2014). What information and knowledge to share with business partners depends on certain contextual factors (Carr and Kaynak 2007). Our study shows that LTO is one of the important contextual factors.

In strategic communication, buyer agents will exchange with supplier agents important information such as future purchasing strategies, innovation and new technologies application strategies, and intentions to implement new supply chain policies (Zhang, Viswanathan, and Henke 2011). If the buyers and suppliers have LTO for their business relationships, such strategic communication will be effective and bring many benefits, for example, enhancing competitive advantages for both parties, reducing transaction costs, and both parties can achieve their goals in the sales and purchasing transactions (Du et al. 2012; Zhang, Viswanathan, and Henke 2011). Therefore, supplier agents tend to cooperate with buyer agents instead of behaving opportunistically.

However, providing important confidential information can cause certain risks to buyers if the buyers and suppliers do not have LTO (Yigitbasioglu 2010). When there is no intention to build a long-term relationship, supplier agents can take advantage of provided information from buyer agents to act opportunistically, such as forcing prices, exaggerating their product values and supply capacities to meet buyers' expectations, or they can make excessive requirements for contract terms, or even ask to change the contract for their own advantages (Chung, Sternquist, and Chen 2006).

In sum, the main results in this study imply that buyer agents' strategic communication generally negatively impacts supplier agents' opportunism; however, buyer agents' strategic communication

needs to be combined with the LTO between buyer and supplier firms to create an effective mechanism for controlling opportunism.

Conclusions

This study aimed to investigate the impact of buyer agents' BSCs and LTO between buyer and supplier firms on supplier agent opportunism, as the mechanisms through which these factors interact. To achieve this, a survey was conducted among 406 purchasing staff in the sales and planning sections of manufacturing companies in Japan. Regression and conditional process analyses were employed to evaluate the main effects of BSCs on opportunism and the interaction effects between BSCs and LTO in managing opportunism. The results reveal that among the three types of BSCs, strategic communication plays a crucial role in helping buyer agents reduce the risk of opportunistic behaviours from supplier agents. Moreover, LTO between the buyer and supplier companies significantly enhanced the effectiveness of BSCs in reducing supplier agents' opportunistic behaviours. Based on these findings, this study makes the following contributions.

Theoretical contributions

First, by examining individual-level opportunism, this study extends TCT, RDT, BST, and the literature on opportunism in inter-firm relationships. The current research trend has emphasised the individual level of analysis, but there is still lack of sufficient studies on individual opportunism (Gelderman, Semeijn, and Verhappen 2020; Hawkins, Lewin, and Amos 2012; Hawkins, Pohlen, and Prybutok 2013; Tangpong, Hung, and Ro 2010). This study answers the call for further research on individual opportunism by investigating the factors influencing individual supplier agents' opportunism. Our findings confirm the need for research on individual opportunism and highlight the importance of individual agents in managing opportunistic behaviours in business exchanges between companies.

Second, this study enriches the literature on factors influencing opportunism by explaining the correlation between BSCs and opportunism. The effects of BSCs in controlling opportunism are fundamental assumptions of TCT and RDT (Williamson 1985; Pfeffer and Salancik 1978), but limited research has provided detailed elaboration on this relationship. Our study complements these theories by operationalising BSCs as buyer agents' professional purchasing knowledge, compromise ability, and strategic communication (Zhang, Viswanathan, and Henke 2011) and testing their effects on supplier agents' opportunism.

Third, this study extends the understanding of the integrated framework in previous studies regarding the interaction effects between organisational and individual factors in managing individual opportunism in business exchanges (Gelderman, Semeijn, and Verhappen 2020; Hawkins, Lewin, and Amos 2012; Hawkins, Pohlen, and Prybutok 2013). In this study, we propose a mechanism where LTO between buyer firms and supplier firms is an organisational factor that can interact with individual factors such as buyer agents' BSCs, to effectively mitigate supplier agents' opportunism. By doing so, this study confirms the direct relationship between LTO and opportunism that was observed in previous studies and contributes to the research field by further explaining the boundary condition of LTO in controlling opportunism.

Implications for practitioners

Our results also suggest three implications for professionals working in sales and purchasing sections and who are involved in inter – firm relationship management in manufacturing companies in Japan. First, buyer agents should develop their strategic communication skills. These skills enable buyer agents to effectively communicate with supplier agents regarding buyer firms' desires and interests in long – term plans. Such communication is critical for reducing supplier agents' opportunistic behaviours. Second, while purchasing professional knowledge and ability to compromise are important, they alone cannot help control supplier agents' opportunistic behaviours without a long – term cooperation orientation between parties. Third, LTO is critical in business exchanges as it helps sustain a healthy cooperative environment where BSCs can effectively control opportunism.

Limitations and future research

It is important to consider the findings in the context of certain limitations. First, the specific types of buyer – supplier relationships in supply chains were not clarified in detail. The dynamics, technology, and product exchanges in distributor – maker and maker – supplier relationships may differ, leading to variations in opportunistic behaviours and the mechanisms to effectively control them. Second, our data only represented the buyer side, without considering potential differences in perceptions about relationship quality among suppliers. Third, our hypotheses were tested using cross – sectional data, limiting our ability to provide detailed explanation of how, when and why BSCs and LTO can result in opportunistic behaviours. Fourth, we considered opportunism as a multidimensional construct, taking into different manifestations of opportunistic behaviours.

Future research can further elaborate on our findings by focusing on specific positions within the buyer – supplier relationship in supply chains, such as distributor – maker, or maker – supplier relationships. Additionally, data collection should be collected from both the buyer and supplier perspectives to mitigate bias and measurement fallacies when assessing dyadic constructs. Moreover, future research can employ experimental methods or develop measurements for each type of opportunism to examine the effects of different spanning capabilities on various opportunism. Such research endeavours can offer valuable insights into how BSCs can help reduce opportunism.

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Appendix Sample profiles

	Characteristics	Number	%
Manufacturing industry	Grocery	46	11.3
	Fibre products	15	3.7
	Pulp, paper	4	1.0
	Chemistry	42	10.3
	Pharmaceuticals	31	7.6
	Petroleum and coal products	9	2.2
	Rubber product	7	1.7
	Glass, earth and stone products	10	2.5
	Steel	5	1.2
	Non-ferrous metal products	15	3.7
	Metal products	17	4.2
	Machine	44	10.8
	Electrical equipment	67	16.6
	Transportation equipment	20	4.9
	Precision mechanical equipment	35	8.6
	Other products	39	9.7
	Total	406	100%
Size of companies (number of employees)	1–10	19	4.7
	11–100	93	22.9
	101–200	35	8.6
	201–300	31	7.6
	301–400	18	4.4
	401–500	12	3.0
	More than 500	198	48.8
Percentage of mutual capital between buyer and supplier company	Total	406	100%
	No capital	323	79.6
	Less than 20%	26	6.4
	20–40%	34	8.4
	40–60%	13	3.2
	60–80%	3	0.7
	More than 80%	7	1.7
Respondents' gender	Total	406	100%
	Men	223	55
	Women	183	45
Respondents' job level	Total	406	100%
	Executives	39	9.7
	Manager	85	20.9
	Section manager	122	30.0
	Unit manager	79	19.5
	General employee	78	19.2
	Other professionals, special jobs, etc.	3	0.7
Respondents' age (years)	Total	406	100%
	15–19	0	0.0
	20–29	11	2.7
	30–39	60	14.8
	40–49	111	27.3
	50–59	155	38.2
	60–69	69	17.0
Respondents' years of experience as buyers	Above 70	0	0.0
	Total	406	100%
	Less than 5	129	31.8
	5–9	68	16.7
	10–14	53	13.1
	15–19	38	9.4
	20–24	42	10.3
	25–29	26	6.4
	30–34	24	5.9
	More than 35	26	6.4
	Total	406	100%